Дано: Т3

$$\xi \sim p(x) = \begin{cases} e^{-\frac{x}{\theta}}, & x \geq 0 \\ 0, & x < 0 \end{cases}, \quad \theta > 0 \qquad n = 3 \qquad \tilde{\theta}_1 = \bar{x}$$

$$\tilde{\theta}_2 = x_{(2)}$$

$$M\xi = \int_{-\infty}^{+\infty} x\, dF(x) = \int_0^{+\infty} \frac{x}{\theta} e^{-\frac{x}{\theta}} dx = -\int_0^{+\infty} x\, de^{-\frac{x}{\theta}} = -xe^{-\frac{x}{\theta}}\Big|_0^{+\infty} + \int_0^{+\infty} e^{-\frac{x}{\theta}} dx$$

$$= \theta\left(-e^{-\frac{x}{\theta}}\right)\Big|_0^{+\infty} = \theta$$

$$M\xi^2 = \int_{-\infty}^{+\infty} x^2 dF(x) = \int_0^{+\infty} \frac{x^2}{\theta} e^{-\frac{x}{\theta}} dx = -\int_0^{+\infty} x^2 d\left(e^{-\frac{x}{\theta}}\right) =$$

$$= -x^2 e^{-\frac{x}{\theta}}\Big|_0^{+\infty} + \frac{\theta}{\theta}\int_0^{+\infty} 2x e^{-\frac{x}{\theta}} dx = 2\theta^2$$

$$D\xi = M\xi^2 - M\xi^2 = \theta^2$$

а) $M\tilde{\theta}_1 = M\bar{x} = M\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n} n \cdot M\xi = \theta$

$\boxed{\Rightarrow \tilde{\theta}_1 \text{ явл. несмещ.}}$

$$M\tilde{\theta}_2 = M x_{(2)}$$

$$x_{(k)} \sim \sum_{i=k}^{n} C_n^i f^i_{(t)} (1 - F_{(t)})^{n-i}$$

$$x_{(2)} \sim \sum_{i=2}^{3} C_3^i F^i_{(t)} (1 - F_{(t)})^{3-i}$$

$$P_{(2)}(x) = \underbrace{\frac{3!}{1!\,1!}}_{6} (F(x))^1 (1 - f(x))^1 \cdot P(x)\, F(x) = 1 - e^{-\frac{x}{\theta}}$$

$$P(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}}$$

$$\Rightarrow P_{(2)}(x) = 6\left(1 - e^{-\frac{x}{\theta}}\right)\left(e^{-\frac{x}{\theta}}\right) \cdot \frac{1}{\theta} e^{-\frac{x}{\theta}} = \frac{6}{\theta} e^{-\frac{2x}{\theta}}\left(1 - e^{-\frac{x}{\theta}}\right)$$

$$M x_{(2)} = \int_{-\infty}^{+\infty} x \cdot P_{(2)}(x)\, dx = \frac{6}{\theta}\int_0^{+\infty} x e^{-\frac{2x}{\theta}}\left(1 - e^{-\frac{x}{\theta}}\right) dx$$

$$W_1 = \int_0^{+\infty} x e^{-\frac{2x}{\theta}} dx = \frac{1}{(2/\theta)^2} = \frac{\theta^2}{4}$$

$$W_2 = \int_0^{\infty} x e^{-\frac{3x}{\theta}} dx = \frac{\theta^2}{9}$$

$$M X_{(2)} = \frac{6}{\theta}\left(\frac{\theta^2}{4} - \frac{\theta^2}{9}\right) = 6\theta\left(\frac{5}{36}\right) = \frac{5}{6}\theta$$

$\Rightarrow$ оц. смещённая

Исправляем:

$$\boxed{\tilde{\theta}_2' = \frac{6}{5} \cdot X_{(2)}} \Rightarrow M\tilde{\theta}_2' = \theta$$

— смещённая

b) Оценка регулярности:

1) $F(x, \theta) = 1 - e^{-\frac{x}{\theta}}$ — непр. дифф. при $\theta > 0$

2) $$\int_{-\infty}^{+\infty} d\,\frac{\partial F(x,\theta)}{\partial\theta} = \int_{-\infty}^{+\infty} d\,\frac{\partial}{\partial\theta}\left(1 - e^{-\frac{x}{\theta}}\right) = \int_{0}^{+\infty} d\left(e^{-\frac{x}{\theta}}\left(-\frac{x}{\theta^2}\right)\right) = 0$$

3) $$I(\theta) = M\left[\left(\frac{\partial \ln p(x,\theta)}{\partial\theta}\right)^2\right] = \int_0^{+\infty}\left(\frac{\partial \ln p(x,\theta)}{\partial\theta}\right)^2 p(x,\theta)\,dx$$

$$\frac{\partial \ln\left(e^{-\frac{x}{\theta}}/\theta\right)}{\partial\theta} = \frac{\partial\left(-\frac{x}{\theta} + \ln\frac{1}{\theta}\right)}{\partial\theta} = \frac{x}{\theta^2} - \theta\cdot\frac{1}{\theta^2} = \frac{x}{\theta^2} - \frac{1}{\theta}$$

$$\int_0^{+\infty}\left(\frac{x}{\theta^2} - \frac{1}{\theta}\right)^2 P(x,\theta)\,dx = \int_0^{+\infty}\left(\frac{x^2}{\theta^4} - 2\frac{x}{\theta^3} + \frac{1}{\theta^2}\right)\underbrace{P(x,\theta)}_{=\frac{e^{-\frac{x}{\theta}}}{\theta}}\,dx$$

$$W_1 = \int_0^{+\infty}\frac{x^2}{\theta^5}e^{-\frac{x}{\theta}}\,dx = \frac{1}{\theta^4}\int_0^{+\infty} 2x e^{-\frac{x}{\theta}}\,dx = W_2 =$$

$$= -\frac{2}{\theta^3}\int_0^{+\infty} x\,d\left(e^{-\frac{x}{\theta}}\right) = \frac{2}{\theta^3}\int_0^{+\infty} e^{-\frac{x}{\theta}}\,dx = \frac{2}{\theta^2}$$

$$W_3 = \frac{1}{\theta^2}\int_0^{+\infty} e^{-\frac{x}{\theta}}\,d\left(\frac{x}{\theta}\right) = \frac{1}{\theta^2}$$

$$\Rightarrow I(\theta) = \frac{1}{\theta^2} > 0$$

Модель регулярна по опр.

$$D\tilde{\theta}_1 = \frac{1}{n^2} \cdot n D\xi = \frac{1}{n}\theta^2 = \frac{\theta^2}{3}$$

$$M X_{(2)}^2 = \int_{-\infty}^{+\infty} x^2 \cdot P_{(2)}(x)\,dx = \int_0^{+\infty} x^2 \cdot \frac{6}{\theta} e^{-\frac{2x}{\theta}} \left(1 - e^{-\frac{3x}{\theta}}\right) dx =$$

$$= \frac{6}{\theta}\int_0^{+\infty} x^2 e^{-\frac{2x}{\theta}}\,dx - \int_0^{+\infty} \frac{6}{\theta} x^2 e^{-\frac{3x}{\theta}}\,dx = 3\int_0^{+\infty} e^{-\frac{2x}{\theta}} 2x\,dx =$$

$$= 2\int_0^{+\infty} e^{-\frac{3x}{\theta}} \cdot 2x\,dx = -3\theta\int_0^{+\infty} x\,d\left(e^{-\frac{2x}{\theta}}\right) + \frac{4\theta}{3}\int_0^{+\infty} x\,d\left(e^{-\frac{3x}{\theta}}\right) =$$

$$= \frac{3}{2}\theta^2 - \frac{4}{9}\theta^2 = \frac{19}{18}\theta^2$$

$$D X_{(2)} = M X_{(2)}^2 - M X_{(2)}^2 = \frac{13}{36}\theta^2$$

$$D\tilde{\theta}_2' = D\left(\frac{6}{5} X_{(2)}\right) = \frac{36}{25} D X_{(2)} = \frac{13}{25}\theta^2$$

По Т. о Достаточном условии регулярности $\tilde{\theta}_1$ и $\tilde{\theta}_2'$ – регулярны

$$D\tilde{\theta}_1 < D\tilde{\theta}_2' \Rightarrow \boxed{\tilde{\theta}_1 \text{ – более эффективна}}$$

c) Т.к. оценки и модель явл. регулярными – исп. нер-во Крамера-Рао:

$$D\tilde{\theta}_1 = \frac{\theta^2}{3} = \frac{1}{n I(\theta)} = \frac{\theta^2}{n}$$

$$\Rightarrow \boxed{\tilde{\theta}_1 \text{ – явл. эффективной}}$$

для $\tilde{\theta}_2'$ тут точно не выпол. усл: $\ln f \tilde{\theta}_0(\vec{x}_n)$ $\forall \theta \in \Theta$

№ Т4.

$\xi \sim P(x) = \frac{\theta}{2}$, $\theta \in (0,1)$ – вер-ть попадания в $(-1, 1)$

$P(\xi=0) = P(\xi=2) = \frac{1-\theta}{2}$

$\cancel{M_\xi = \int_{-1}^{1} \frac{\theta}{2} x\,dx = \frac{\theta}{2}\frac{x^2}{2}\Big|_{-1}^{1} =}$

$M_\xi = \underbrace{\int_{-1}^{1} \frac{\theta}{2} x\,dx}_{=0} + \underbrace{0 \cdot \frac{1-\theta}{2}}_{=0} + 2 \cdot \frac{1-\theta}{2} = 1-\theta$

$M_{\xi^2} = \int_{-1}^{1} \frac{\theta}{2} x^2\,dx + 4 \cdot \frac{1-\theta}{2} = 2 - \frac{5}{3}\theta$

$D_\xi = 2 - \frac{5}{3}\theta - (1-\theta)^2 = 2 - \frac{5}{3}\theta - 1 + 2\theta - \theta^2 = 1 + \frac{1}{3}\theta - \theta^2$

а) $\tilde{\theta}_1$ – ОММ

$\bar{x} = 1 - \theta \Rightarrow \boxed{\tilde{\theta}_1 = 1 - \bar{x}}$

$\tilde{\theta}_2$ – ОМП

$L(\theta) = \prod_{i=1}^{n} p(x_i, \theta)$

кол-во „2" = m, кол-во „0" = k

соответственно других чисел: $n-m-k$

$L(\theta) = \left(\frac{1-\theta}{2}\right)^{m+k} \cdot \left(\frac{\theta}{2}\right)^{n-m-k} = \left(\frac{1-\theta}{2}\right)^{t} \left(\frac{\theta}{2}\right)^{n-t}$ # $m+k=t$

$\ln L(\theta) = t \ln\left(\frac{1-\theta}{2}\right) + (n-t) \ln \frac{\theta}{2}$

Найдём $\sup_{(0,1)} \ln L(\theta)$ Т.к. мн-во открытое

$\Rightarrow \frac{\partial}{\partial\theta} \ln L(\theta) = -\frac{2t}{2(1-\theta)} + \frac{n-t}{\theta} \cdot \frac{1}{2} = \frac{n-t}{\theta} - \frac{t}{1-\theta} = 0$

$(n-t)(1-\theta) = t\theta$

$n - t - n\theta + t\theta = t\theta \Rightarrow \theta = \frac{n-t}{n} = 1-\nu$

$$\frac{\partial^2 (\ln L(\theta))}{\partial \theta^2} = -\frac{(n-t)}{\theta^2} - \frac{t}{(1-\theta)^2} < 0$$

$\Rightarrow \theta$ – макс

$\boxed{\tilde{\theta}_2 = 1-\nu}$, где ν – частота появл. 0 и 2.

b) $M\tilde{\theta}_1 = 1 - \frac{1}{n} \cdot n \cdot M\xi = 1-(1-\theta) = \theta$

$\longrightarrow$ несмещ.

$$D\tilde{\theta}_1 = D\bar{x} = \frac{1}{n^2} \cdot nD\xi = \frac{1}{n} - \left(1 + \frac{1}{3}\theta - \theta^2\right) \xrightarrow[n\to\infty]{} 0$$

$\Rightarrow$ $\boxed{\tilde{\theta}_1 \text{ – состоят по достаточ. усл.}}$

$M\tilde{\theta}_2 = M(1-\nu) = \theta$ – несмещ.

$M\nu = P(\xi = 0 \text{ или } \xi = 2) = 1-\theta$

$$D\tilde{\theta}_2 = D(1-\nu) = D\nu = \frac{p(1-p)}{n} = \frac{\theta(1-\theta)}{n} \xrightarrow[n\to\infty]{} 0$$

$\Rightarrow$ $\boxed{\tilde{\theta}_2 \text{ – состоят. по достаточ. усл.}}$

c) 1) $P(x,\theta) = \frac{\theta}{2}$, $P_k = \frac{1-\theta}{2}$ – дифф. по θ на $(0,1)$

2) $\frac{\partial}{\partial\theta}\int_{-\infty}^{+\infty} dF = 0$ $\quad \int_{-1}^{1} \frac{1}{2}dx + 2\left(-\frac{1}{2}\right) = 0$

3) $$I(\theta) = \int_{-1}^{1}\left(\frac{\partial \ln\frac{\theta}{2}}{\partial\theta}\right)^2 \frac{\theta}{2}dx + 2\left(\frac{\partial\left(\frac{1-\theta}{2}\right)}{\partial\theta}\right)^2 \frac{1-\theta}{2} =$$

$$= \theta\left(\frac{2}{\theta}\cdot\frac{1}{2}\right)^2 + (1-\theta)\left(\frac{2}{1-\theta}\cdot\left(-\frac{1}{2}\right)\right)^2 =$$

$= \frac{1}{\theta} + \frac{1}{1-\theta} = \frac{1}{\theta(1-\theta)}$ – непрерыв. и >0 на $(0,1)$

$\Rightarrow$ модель <u>регулярна</u> по определению

$D\tilde{\theta}_1$ и $D\tilde{\theta}_2$ - огранич. на компакте из $(0,1)$

Тогда $\tilde{\theta}_1$ и $\tilde{\theta}_2$ - регулярны по достат. усл.

$\Rightarrow$ Нер-во Крамера-Рао применимо.

$$D\tilde{\theta}_1 = \frac{1}{n}\left(1+\frac{1}{3}\theta-\theta^2\right) > \frac{1}{n}(\theta)\cdot(1-\theta) = \frac{1}{n}(\theta-\theta^2) = \frac{1}{n\cdot I(\theta)}$$

$\Rightarrow$ не можем делать выводов об эффективности $D\tilde{\theta}_1$

$$D\tilde{\theta}_2 = \frac{1}{n}\theta(1-\theta) = \frac{1}{n\cdot I(\theta)}$$ $\Rightarrow$ выполнено достат. усл. эфф.

$\Rightarrow$ $\tilde{\theta}_2$ - эффективная оценка.